Examiner only

**3.** An artificial gene coding for human growth hormone (HGH), was synthesised in a laboratory. The synthesised gene was used to genetically modify *E. coli* bacteria, which then produced HGH. The synthesised HGH was identical to natural HGH in size and biological activity.

(a) The human genome project determined the base pairs that make up human DNA. Explain how the results from the human genome project have allowed the synthesis of the artificial gene to become possible. [1]

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(b) State **three** advantages of using an artificially synthesised gene rather than extracting the gene from the human genome. [3]

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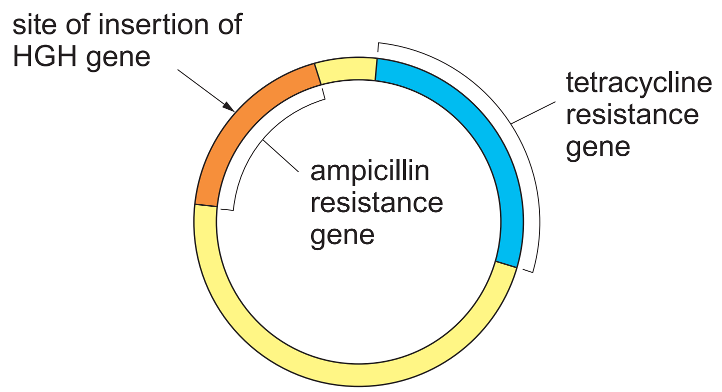
The plasmid used to genetically modify the *E. coli* is shown in **Image 3.1**. It also shows the positions of two antibiotic resistance genes and the site of insertion of the HGH gene.

**Image 3.1**




(c) Name the **two** types of enzymes needed for the successful insertion of the human growth hormone gene and state their function in the process. [2]

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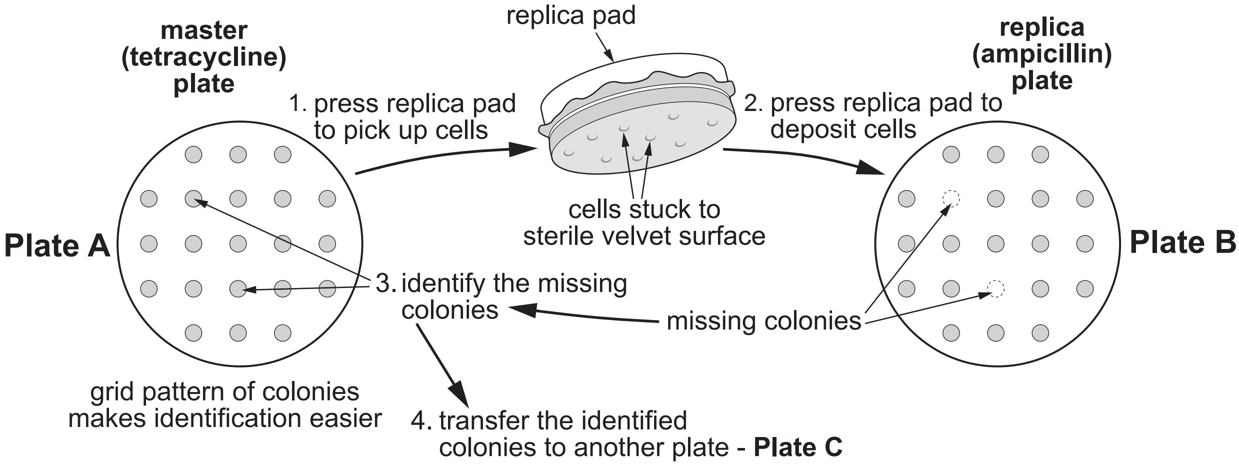
To test that the plasmid has transferred successfully, the bacteria were grown on separate agar plates containing the two antibiotics. The bacteria were initially grown on agar containing tetracycline and the colonies were then transferred, using a replica pad, to agar containing ampicillin as shown in **Image 3.2**.

**Image 3.2**




(d) (i) Explain the need for **Plate A** in the process. [1]

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(ii) Explain the results shown on **Plate B**. [3]

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(iii) Suggest **one** vital property of the cells on **Plate C** which would need to be confirmed prior to their use in the production of HGH. [1]

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(e) Each *E. coli* cell transformed with this recombinant plasmid produced $2.9 \times 10^6$ molecules of HGH. If there are $1 \times 10^8$ bacteria in the culture vessel, calculate the number of HGH molecules being produced by the cells. **Give your answer in standard form**. [2]

Number of HGH molecules = ..............................................

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